

# EEE4114F Final Exam

June 2025

Duration: 3 hours

Total: 80 marks

Empl ID: \_\_\_\_\_

## Instructions:

- This is a closed-book exam.
- There are two parts to this exam.
- **Part A** has 5 questions totalling 40 marks. You must answer all of them.
- **Part B** has 3 questions totalling 40 marks. You must answer all of them.
- Both parts can be answered in the space provided on the exam script, and you can write on the back of the printed pages if required — but please specify where you do so. Answer books will be available should you need additional writing space.
- You must use a pen to complete the exam — pencil will only be marked for drawings. Values indicated on drawings must be written in pen. Please show all of your working clearly — method and approach matter.
- A table of standard Fourier transform and z-transform pairs is provided separately.
- You have 3 hours.

| Question | Marks | Grade |
|----------|-------|-------|
| 1        | 10    |       |
| 2        | 5     |       |
| 3        | 5     |       |
| 4        | 10    |       |
| 5        | 10    |       |
| 6        | 13    |       |
| 7        | 17    |       |
| 8        | 10    |       |
| Total:   | 80    |       |

## Part A: Digital Signal Processing

### Question 1 [10 marks]

Consider the sequence

$$x[n] = \delta[n+1] - 2\delta[n] + \delta[n-1].$$

Evaluate and sketch each of the following signals over a sensible range:

1.  $y_1[n] = x[n-2]$

2.  $y_2[n] = -x[-\frac{1}{2}n]$

3.  $y_3[n] = x[n+2]\delta[n+1]$

4.  $y_4[n] = x[n+1] * \delta[n-1]$

5.  $|X(e^{j\omega})|$  (i.e., the magnitude of the DTFT of  $x[n]$ )

**Question 2** [5 marks]

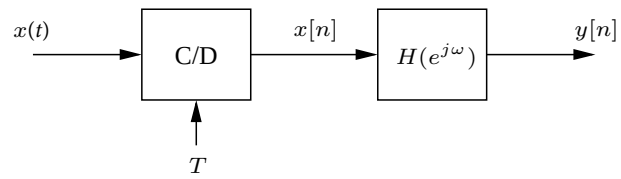
Consider the following discrete-time signal:

$$x[n] = \begin{cases} 4 - n & 0 \leq n \leq 3 \\ 1 & n \geq 4 \\ 0 & \text{otherwise.} \end{cases}$$

Determine the z-transform  $X(z)$  of  $x[n]$  such that  $X(z)$  is a ratio of polynomials in  $z^{-1}$ .

**Question 3** [5 marks]

Consider the system below

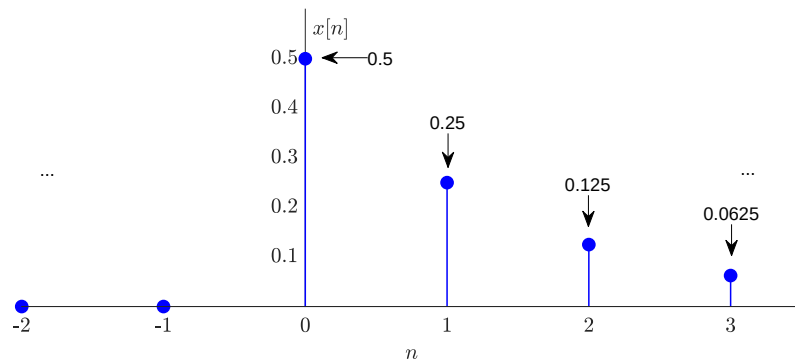
where  $T = 0.01$  s and

$$H(e^{j\omega}) = \begin{cases} 1 & |\omega| \leq 0.5\pi \\ 0 & \text{otherwise} \end{cases}$$

Find the output  $y[n]$  if the input is  $x(t) = \cos(20\pi t) + \cos(80\pi t)$ .

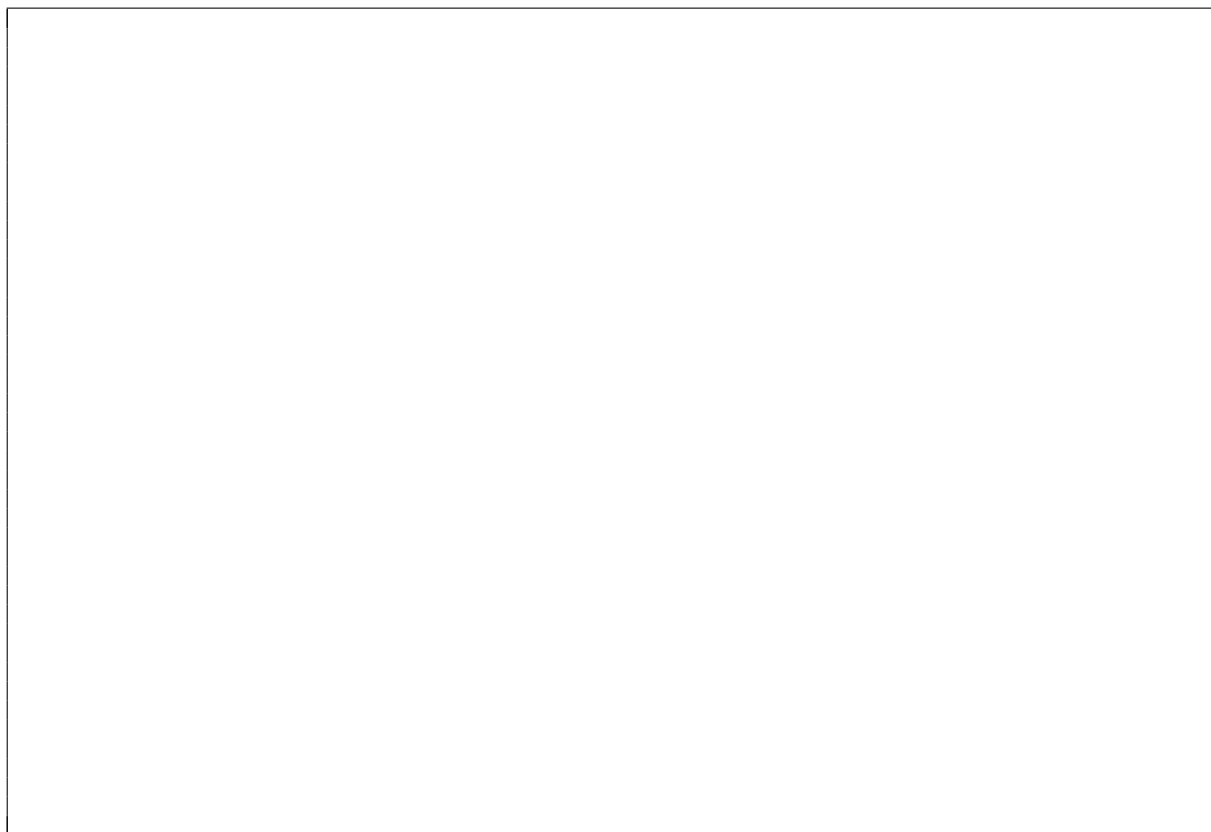
**Question 4** [10 marks]

Consider the aperiodic, infinitely-long decaying exponential sequence  $x[n]$  below:

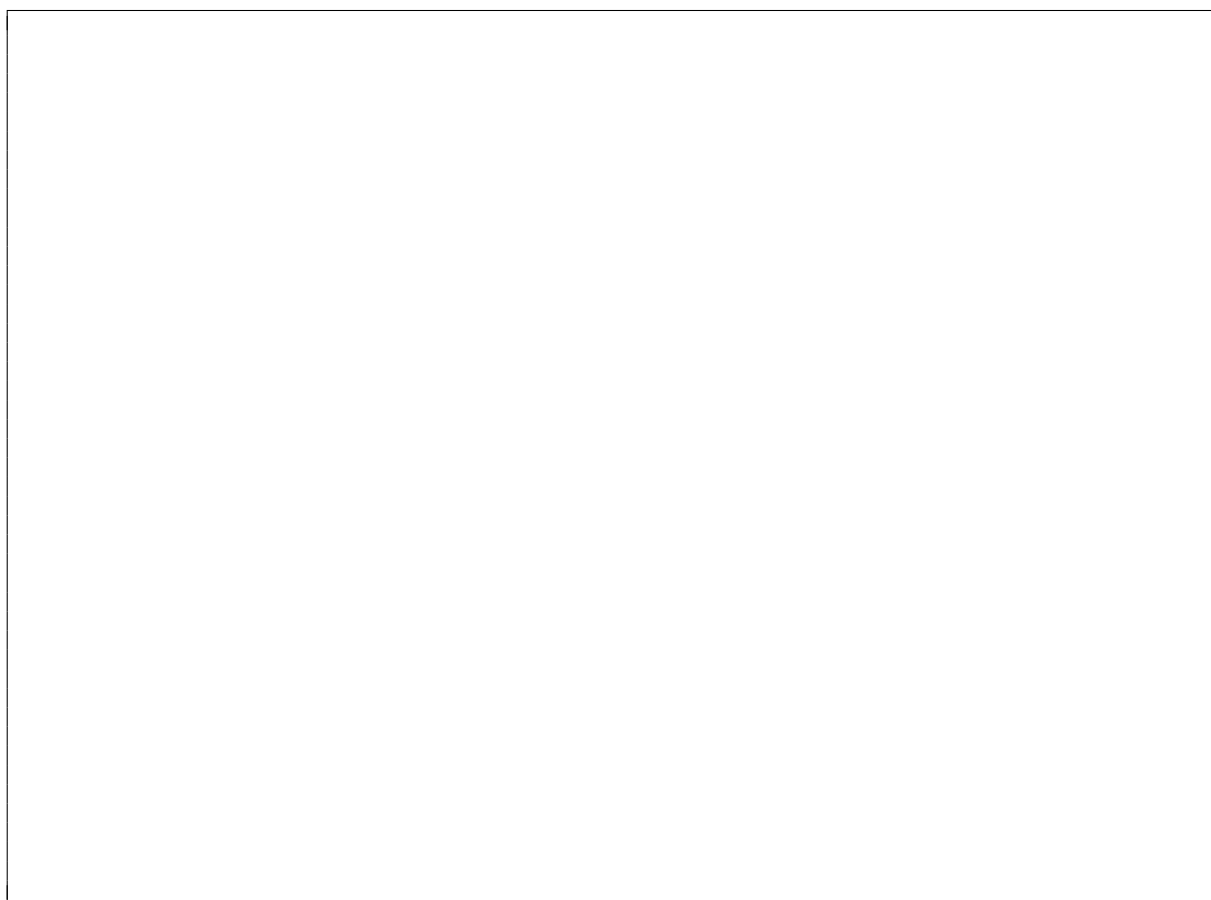


1. Provide an analytical expression for  $x[n]$ .

2. Calculate the 4-point DFT  $X[k]$  of the signal  $x[n]$ .

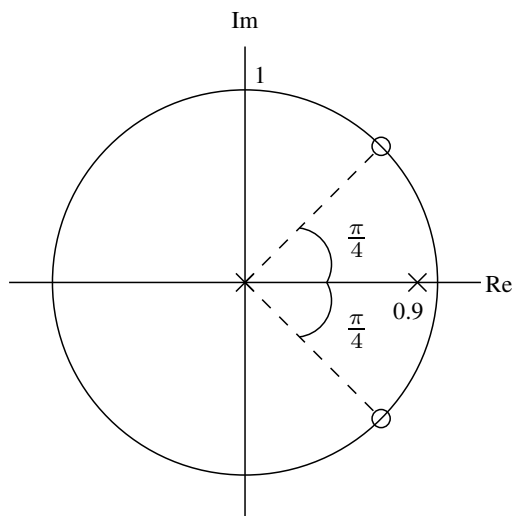


3. Sketch the signal  $y[n] = x[((n+3))_4]$ .



**Question 5** [10 marks]

The following figure indicates a system with two poles and two zeros:



1. Determine the transfer function  $H(z)$  of the system if

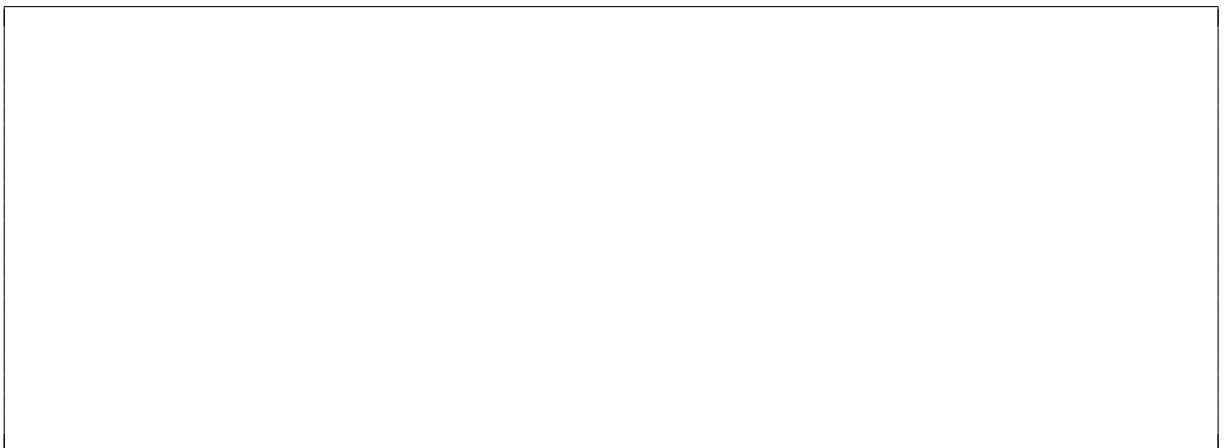
$$H(z)|_{z=1} = 1.$$



2. Provide a rough sketch of the magnitude of the frequency response of the system over the range  $0 \leq \omega \leq 2\pi$ .



3. Comment on the system's stability and causality for each possible ROC.



**END OF PART A**

## Part B: Machine Learning

### Question 6: Fundamental Concepts [13 marks]

Suppose you are trying to develop a system that can estimate the yield of various agricultural crops such as wheat and cotton. Data is collected from a few farms across the world. Data was collected from five regions in total: Central USA, East Africa, North India, South USA, and Southern Africa. Your system is intended to generalize to farms that are not part of the dataset, but are still located within those five regions.

The data are summarized in Table 1

| Variable Name | Type        | Units       |
|---------------|-------------|-------------|
| farm_id       | Integer     | -           |
| region        | Categorical | -           |
| soil_moisture | Continuous  | %           |
| soil_pH       | Continuous  | pH          |
| crop_type     | Categorical | -           |
| temperature   | Continuous  | $^{\circ}C$ |
| yield         | Continuous  | kG/ha       |
| rainfall      | Continuous  | mm          |
| farm_owner    | Categorical | -           |

Table 1: A summary of the collected data

1. Are there any input features you would consider unnecessary for crop yield estimation? Explain your answer. (2)

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

2. Suppose you split your data with a single holdout method such that farms from Southern Africa are not present in the training data, and only in the test data. Would this be a reasonable resampling approach? (2)

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

3. Suppose you keep the test split of data from question 6.2 as it was, but now further split the training data using cross-validation to produce multiple, independent validation sets. Would this be a reasonable resampling approach?

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

4. Suppose you decide to try an alternative resampling approach by randomly splitting the original dataset with a single holdout method into training and test sets, however, this time you sample with replacement (examples can be reused) to create a larger dataset. Would this be a reasonable resampling approach?

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on its right side, suggesting it's resting on a surface.

5. You manage to train three different models with an appropriate resampling method to achieve the results (3) shown in Figure 1. Which of these models would benefit (increase in their respective  $r^2$  scores on a separate test set) from retraining on a combined training and validation set? Explain your answer.

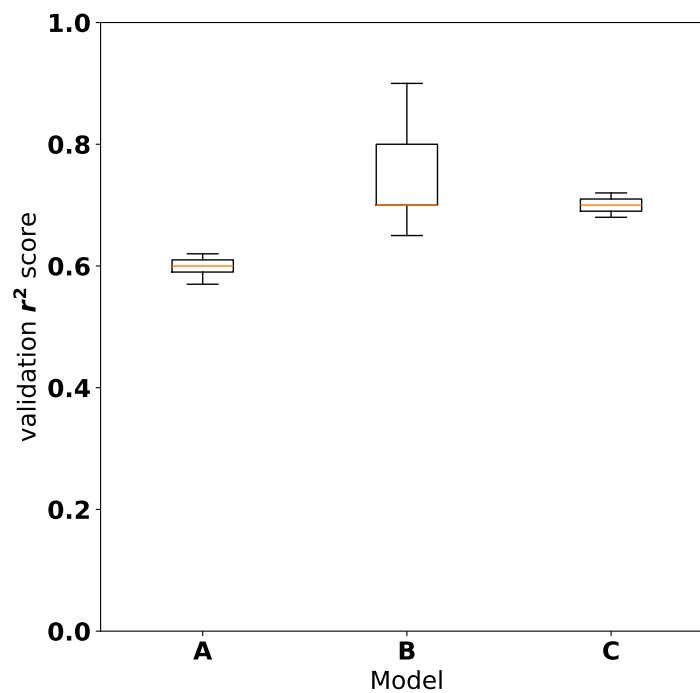


Figure 1: A box-whisker plot of the  $r^2$  scores after cross-validation of the model.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(2)

[illegible]

**Question 7: Neural Networks** [17 marks]

Suppose you have a dataset,  $\mathcal{D}$ , for the problem in Q6. You decide to train an artificial neural network to solve the problem and you also decide to limit the number of input features. The hidden layer uses relu activation functions and the output uses a linear activation.

The initial neural network parameters are:

$$\mathbf{W}^1 = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \\ 1 & 1 \end{bmatrix}, \mathbf{W}^2 = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}, \mathbf{b}^1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, b^2 = 0$$

The following values are determined after the first forward pass:

$$\mathbf{z}^1 = \begin{bmatrix} 1.2 \\ 1.25 \end{bmatrix}, \mathbf{z}^2 = [1.85]$$

The cost function is the mean square error  $J = \frac{1}{2}(\hat{y} - y)^2$

Answer the following questions:

1. How many input features were used, considering the shape of the neural network parameters? (1)  
1. \_\_\_\_\_
2. How many neurons are there in the hidden layer of this neural network? (1)  
2. \_\_\_\_\_
3. Draw a graphical representation of the neural network and label all input features, model parameters, net inputs, and activations. You do not need to include numerical values, just the symbolic representation e.g.  $x_1$ ,  $w_{11}^1$ ,  $z_1^1$ ,  $a_1^1$  etc. (2)

4. Write out an expression for  $\frac{\partial J}{\partial w_{22}^1}$  using chain rule. (2)

5. Calculate the value for  $\frac{\partial J}{\partial w_{22}^1}$  if  $x_2 = 0.6$  and  $y = 1.7$  for the first forward pass (6)

6. What type of layer in a convolutional neural network helps make them invariant to translational shifts in the data? Why? (2)

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

7. Suppose you have a 2-dimensional input with dimensions of  $32 \times 32$ . If you had a 2D convolutional neural network that has a convolutional layer with 5,  $6 \times 6$ , kernels. How many model parameters are there for that layer assuming bias terms are included? How does that compare to a fully-connected neural network with just a single hidden neuron (assuming the input data is flattened to a 1D vector)?





**Question 8: Reinforcement Learning [10 marks]**

Suppose an agent is placed in an environment as shown in Figure 2a with Q-values in Figure 2b that has been developed through training an agent in a virtual environment. The agent can take actions from the following action-space  $\mathcal{A} = \{\text{North, East, South, West}\}$  which causes it to move in the corresponding direction. These can be abbreviated N,E,S,W and the position of the Q value within a cell corresponds to the direction of the action. N is  $\uparrow$ , E is  $\rightarrow$ , S is  $\downarrow$ , W is  $\leftarrow$ .  $s_3$  and  $s_7$  are the terminal states, and offer reward values of +1 and +10 respectively.

|       |       |             |       |              |  |
|-------|-------|-------------|-------|--------------|--|
|       |       | $s_2$       | $s_4$ | $s_6$        |  |
| $s_0$ | $s_1$ | $s_3$<br>+1 | $s_5$ | $s_7$<br>+10 |  |

(a) The agent's environment. Rewards are shown if they are non-zero.

|     |      |     |      |     |     |
|-----|------|-----|------|-----|-----|
|     |      | 0.0 | 0.0  | 0.0 |     |
|     |      | 0.0 | 0.64 | 0.0 | 1.2 |
|     |      | 0.0 | 0.5  | 4.5 |     |
| 0.0 | 0.26 | 0.0 | 0.0  | 0.0 | 0.0 |
| 0.0 | 0.4  | 0.0 | 0.5  | 0.0 | 0.0 |
| 0.0 | 0.0  | 0.0 | 0.0  | 0.0 | 0.0 |

(b) The Q-values after an arbitrary number of episodes.

Figure 2: The environment and Q-values for a virtual agent

The Q-values are updated according to the following learning rule:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [R(s') + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

Where  $\alpha = 0.5$  and  $\gamma = 0.8$ . Answer the following questions.

- Starting in the state,  $s_2$ , write down the sequence of states, actions, and rewards in the form  $(s_t, a_t, r_{t+1}) \rightarrow (s_{t+1}, a_{t+1}, r_{t+2})$  the agent will experience if it is using a greedy action policy,  $\pi(s) = \arg \max_a q_\pi(s, a)$ , to select actions based on the Q-values in Figure 2b. (2)

---



---

- Given the Q-table in Figure 2b, would an agent selecting actions from a greedy policy and starting in  $s_0$  behave optimally? (2)

---



---



---



---



---



---



---



---



---



---



---



---



---



---



---



---



---



---



---



---



---



---

3. Would training the model using an  $\epsilon$ -greedy policy be beneficial? Refer to the Q-values to provide an example of where this is beneficial. (2)

---

---

---

---

---

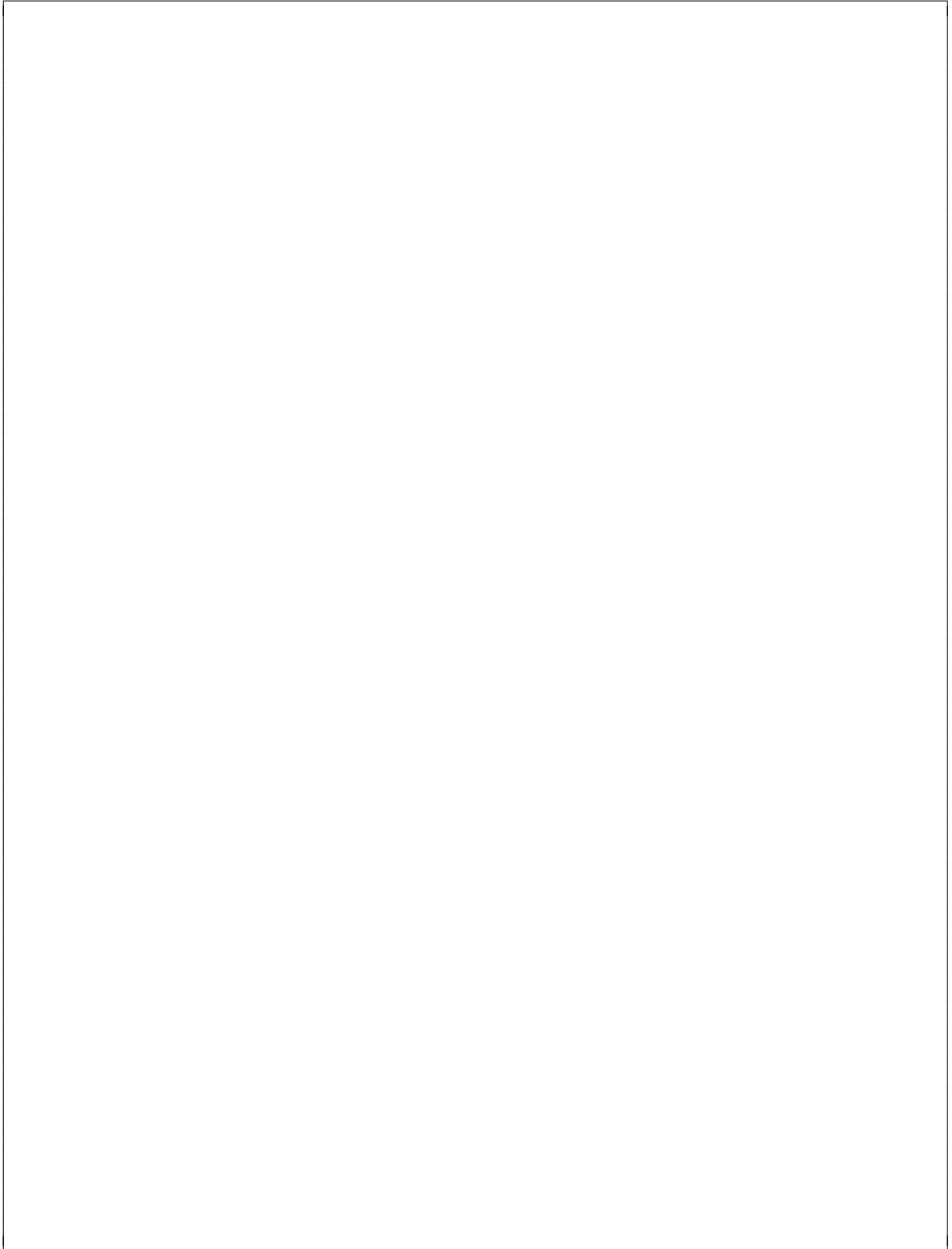
---

---

---

---

4. If the agent starts in state,  $s_0$ , and chooses to go E, N, E, S which Q-values would be updated, and what would their updated values be? (4)



**END OF PART B**